

Learning Human-Grounded Interaction Strategies for Interactive Agents through Action-Mediated Language Policies

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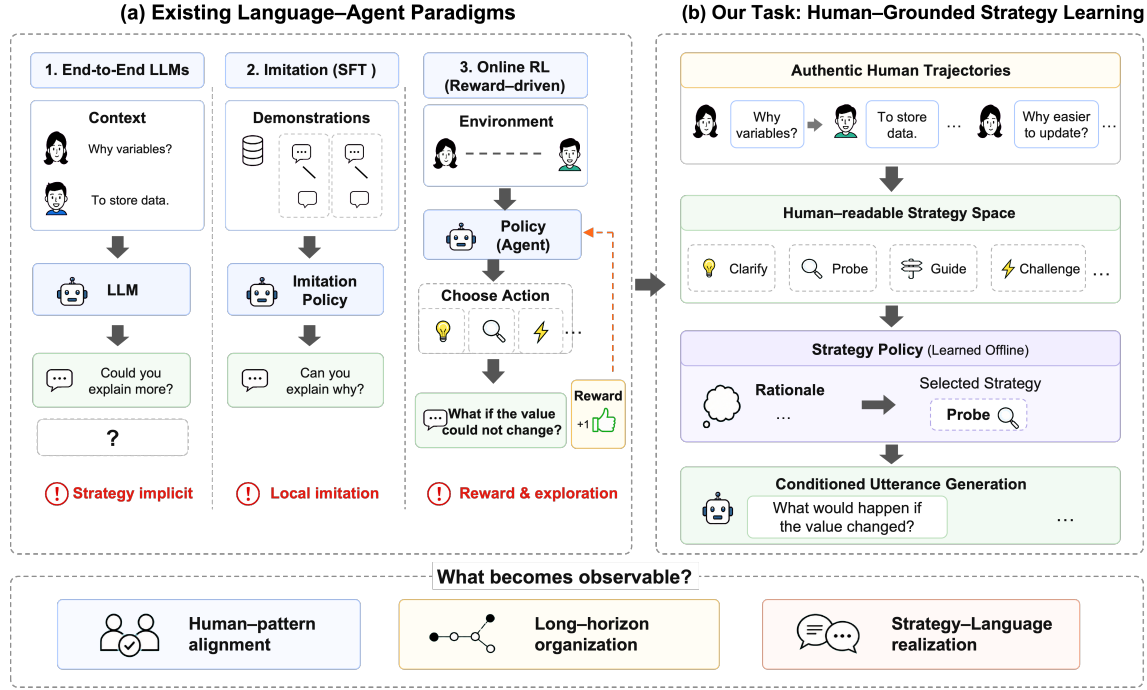


Fig. 1. **Action mediation makes interaction strategy explicit and inspectable.** End-to-end generation leaves strategy implicit, supervised imitation emphasizes local reproduction, and online reinforcement learning depends on reward design and exploration. LA-IQL instead learns offline from authentic human trajectories over a human-readable strategy space, exposes a rationale and selected strategy, and conditions utterance generation on that choice. This explicit boundary makes human-pattern alignment, long-horizon organization, and strategy-language realization separately observable.

Interactive agents can generate fluent responses, yet the interaction strategy behind each response typically remains implicit in end-to-end language generation. This makes strategic choices—such as probing, guiding, challenging, or building on prior contributions—difficult to learn from human interaction, inspect, and coordinate across turns. We introduce Language-Action Inverse Q-Learning (LA-IQL), an action-mediated framework that learns offline from authentic classroom trajectories without online exploration or a hand-designed scalar reward. LA-IQL separates rationale generation, selection over human-readable interaction acts, and language realization. On held-out K-12 lessons, the evaluated TD-regularized checkpoint exhibits less recurrent behavior than its no-TD counterpart while maintaining similar local human-pattern alignment. A metric-matched Markov control shows that strong local alignment can arise without dialogue context, while semantic interventions indicate context-sensitive policy behavior. Direct selection better matches held-out human action patterns, whereas proposal-constrained re-ranking reduces recurrence, revealing an alignment–repetition trade-off. Action mediation thus makes long-horizon interaction strategies explicitly inspectable beyond surface fluency.

CCS Concepts: • **Human-centered computing** → **Human computer interaction (HCI)**; • **Computing methodologies** → *Reinforcement learning*.

Additional Key Words and Phrases: human-grounded interaction, interactive agents, action-mediated policy, offline imitation learning, dialogue strategy

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1 Introduction

Interactive agents have become woven into the fabric of everyday life. Voice assistants field our questions, tutoring systems guide learners through exercises, customer service bots resolve issues, and companion agents provide social engagement. Across these diverse settings, dialogue serves as the fundamental interface through which agents interpret human intent, negotiate meaning, and accomplish tasks [31, 44]. The quality of this dialogic interaction increasingly determines whether these systems feel helpful, natural, and engaging, or mechanical, rigid, and unsatisfying. These gaps between conversational capability and user expectations are long-standing concerns in HCI [7, 26].

Yet a persistent gap separates how current agents converse from how people naturally interact. Most contemporary dialogue agents excel at answering questions and executing direct commands, but struggle with the broader interactional repertoire that characterizes effective human communication. Skilled human interlocutors do far more than respond: they ask clarifying questions, probe incomplete answers, challenge assumptions, offer guidance, build on earlier contributions, and acknowledge and take up what others have said. These interaction-level moves—distinct from the propositional content of any single utterance—shape whether conversations feel dynamic and responsive or flat and repetitive. When agents lack these capacities, interactions become predictable exchanges of information: the agent waits for a question, produces an answer, and waits again. This works for simple queries but fails precisely in the rich, multi-turn, mixed-initiative settings where interactive agents hold the most promise [13]. This gap is central to proactive conversational AI, which treats anticipating needs, initiating useful moves, and coordinating an interaction as capabilities beyond reactive answering [8].

The root of this limitation is architectural. Current interactive agents typically generate dialogue end to end, which leaves the interaction strategy behind each response implicit in the generated text [1, 18, 31, 44]. A locally plausible utterance may clarify, challenge, guide, repeat an earlier pattern, or cede initiative, but these strategic choices are never explicitly represented, evaluated, or controlled. Individually fluent turns can accumulate into repetitive or poorly coordinated interaction [21, 47]. Behavioral cloning and supervised fine-tuning provide direct ways to imitate observed behavior, but primarily optimize local action or token likelihood without explicitly modeling how interaction choices unfold over time. Online reinforcement learning can optimize sequential behavior, but requires exploration that may be costly or unsafe when interaction involves people or high-stakes environments [22, 30, 36]. Figure 1 previews how an explicit action boundary makes these otherwise latent strategy decisions observable.

This motivates a central question: *how can interactive agents learn richer, more human-like dialogue strategies from authentic human interaction, while keeping strategy selection separate enough from language realization to be inspected, evaluated, and improved?* We argue that the path forward requires making interaction strategy an explicit, learnable, and inspectable object, not a hidden byproduct of text generation. This means separating what move to make—clarify,

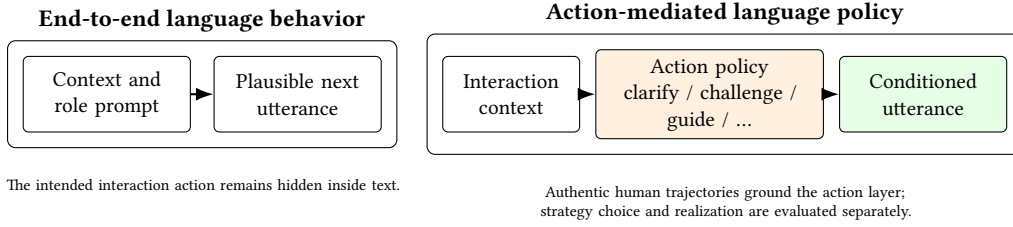


Fig. 2. Human-grounded, action-mediated interaction. LA-IQL learns a human-readable strategy layer from authentic interaction trajectories and separates strategy selection from language realization. This boundary makes behavioral patterns traceable while exposing a realization risk when the final utterance does not express the selected action.

challenge, guide, or acknowledge—from how to word it, so that each can be learned from human demonstration and evaluated on its own terms.

We study this question through a human-grounded formulation in which the action repertoire, training trajectories, and evaluation references are derived from observed human interaction. Here, *human-grounded* refers specifically to the source of the interaction structure: the action space is defined by an established cognitive-act annotation scheme [39], and training comes from authentic transcripts. It does not imply that generated dialogue is globally human-like, natural, contextually appropriate, or effective for people.

Our central contribution is an action-mediated paradigm in which a language policy first chooses a discrete, human-readable interaction act and then realizes that choice in language. This separates the source of a strategy—human demonstrations—from the wording of any individual turn. It also creates a traceable boundary at which strategy learning, temporal organization, control allocation, and the eventual language realization can be evaluated as related but distinct objects. Learning from fixed trajectories avoids online exploration and a hand-designed scalar reward [22, 30, 36].

We investigate this problem through classroom dialogue, a testbed containing frequent, fine-grained interaction moves. Teacher–student exchanges are multi-role and open-ended, combining questioning, uptake, elaboration, challenge, guidance, and synthesis rather than following a fixed task script. Authentic transcripts pair this richness with an established cognitive-act annotation scheme, which provides an observable operationalization of interaction strategies [39]. Our aim is to study how strategy-rich human interaction can ground an interactive agent’s conversational policy.

We propose **Language-Action Inverse Q-Learning (LA-IQL)**, an offline framework configured by a domain-specific action space and human demonstrations. LA-IQL learns action trajectories with rationale-conditioned offline inverse soft Q-learning over fixed transitions [11, 43], selects an interaction act from the learned policy, and delegates its language realization to a role model. We evaluate whether generated trajectories reproduce held-out human action patterns, whether that alignment coexists with repetitive behavior, and whether rationales carry action-relevant information. We additionally evaluate how action-control allocation relates to observable knowledge introduction, connection, and development across the resulting long-horizon dialogues. Across 250 evidence-grounded knowledge graphs, Direct selection introduces fewer unique entities than Candidate while retaining nearly identical relational degree and late-stage relation growth. This reveals a more compact rather than uniformly richer observable knowledge-structure pattern. Figure 2 contrasts this action-mediated interface with an end-to-end language policy.

Our contributions are threefold:

- An action-mediated formulation for human-grounded interaction-strategy learning that separates explicit strategy selection from language realization, enabling interactive agents to learn not just what to say but how to interact.
- LA-IQL, which combines rationale-conditioned action prediction with offline inverse Q-learning over fixed human interaction trajectories, learning sequential interaction policies without online exploration or hand-crafted rewards.
- A long-horizon evaluation protocol for examining alignment with human action patterns, recurrence, context sensitivity, associations between rationales and actions, language realization, and knowledge structure. This reveals behavioral differences that local similarity or fluent text alone can obscure.

By making interaction strategy explicit and learnable from human demonstration, this work takes a step toward interactive agents that do not merely answer questions, but engage in the rich, adaptive, and genuinely collaborative dialogue that makes human conversation effective.

2 Related Work

2.1 Interaction-Strategy Learning in Pedagogical Dialogue

End-to-end dialogue generation has progressed from sequence-to-sequence and hierarchical models to pretrained Transformers [37, 40, 42]. Role prompting and multi-agent simulation now support extended social, task, and classroom interactions [31, 44, 46], but fluent role realization does not by itself expose or control the pedagogical decision that produced an utterance. Recent educational systems therefore introduce intermediate instructional structure. BIPED derives fine-grained tutor and student dialogue acts from human tutoring data and first predicts a tutor act before generating its response [20]. Even with strong LLMs, predicting the next tutor strategy remains difficult, while tutor strategies are informative about subsequent student outcomes [15]; this motivates treating strategy prediction as an explicit problem rather than assuming it is recovered by fluent generation. StratL steers an LLM through a pre-specified multi-turn tutoring plan represented as a transition graph [32], while ScaffoldLM maintains a stepwise plan, inferred learner state, and assessment-driven memory to select tutoring actions across turns [24]. Together, these systems show a shift from unconstrained response generation toward explicit pedagogical planning and action selection. LA-IQL shares this separation of decision and realization, but learns a policy over human-annotated cognitive interaction acts from fixed classroom trajectories rather than prescribing a tutoring plan or inferring a task-specific solution path.

This separation also connects educational dialogue to dialogue-policy learning. Task-oriented dialogue is commonly formulated as sequential decision making, with abstract actions selected from dialogue states and policies optimized by reinforcement learning [22, 23]. In educational dialogue, recent online RL aligns tutors through simulated student-tutor interaction and an explicit trade-off between pedagogical support and task success [10]. Such simulation-based optimization offers controllable objectives but depends on the validity of the simulator and reward. Classical inverse reinforcement learning instead infers objectives from demonstrated behavior [5, 29, 48]. Offline alternatives avoid new interaction: human-centric offline dialogue training learns from logged conversations [17], CHAI ranks language-model candidates with a learned Q-function [41], IQ-Learn recovers a policy from expert demonstrations [11], and Wulfmeier et al. [43] cast inverse soft Q-learning for language as MLE with temporal-difference regularization. LA-IQL adapts this last objective from token-level imitation to a 31-class teacher-student action policy (two roles $\times$ 15 cognitive-act symbols + END) grounded in authentic classroom trajectories [2, 39].

Action-conditioned generation is an established design, as illustrated by BIPED, and LA-IQL builds on an existing inverse soft Q-learning objective. Our contribution connects an action-mediated formulation, offline sequential

policy learning over human-readable interaction acts, and long-horizon diagnostic evaluation. Specifically, rationale-conditioned action prediction and action-level TD regularization learn a policy from fixed human trajectories, while a separate role model realizes its selected acts. The evaluation then examines local human-pattern alignment alongside recurrence, context sensitivity, and expressed dialogue structure. This combination makes it possible to examine whether reproducing local human action patterns also coincides with low long-horizon recurrence, without online exploration or a hand-designed scalar reward.

2.2 Conversational Agency in Human–AI Interaction

HCI research treats conversational agency as more than response quality: an agent must decide when and how to contribute, how much control to assume, and how its behavior remains legible to people [3, 7, 13]. The Inner Thoughts framework separates covert thought formation from overt participation to support proactive conversation [25]. Studies of shared autonomy find non-monotonic trade-offs: delegating communication can aid multitasking while creating discomfort, whereas shared control can impose monitoring costs [6]. In group conversation, users also resist agents that dominate and seek controls over when, what, and where an agent contributes [14]. These works frame initiative as an allocation of control rather than a uniformly desirable increase in autonomy. Recent CHI work further examines how human and AI agency are understood when a chatbot appears to pursue its own conversational agenda [45]. Our allocation boundary is internal to the generated interaction: we compare a learned policy that selects from the full action space with proposal-constrained re-ranking in which the role model first limits the available actions. This does not test user control or establish a preferred autonomy level; it makes the policy–role-model division inspectable at the action layer.

2.3 Evaluating Interactive Behavior in Pedagogical Dialogue

Evaluation must likewise distinguish structural fidelity from pedagogical quality. MRBench provides human annotations over eight learning-science-based dimensions of tutor responses [28]. ReWIRED annotates fine-grained teaching acts in expert explanatory dialogue and shows both that few-shot LLMs struggle to recover these acts and that functional explanation quality requires evaluation beyond static structural metrics [4]. EducationQ instead evaluates teaching in dynamic multi-agent scenarios and finds that general model capability does not map monotonically to teaching effectiveness [38]. Knowledge-graph representations can provide a semantic layer between dialogue utterances and grounded information items [35], but they do not by themselves establish pedagogical value or learning.

The validity of simulated learners is a separate concern. Teachers report that prompted LLM students can be unnaturally attentive, linguistically complex, and logically inconsistent [27]; a broader 2026 benchmark likewise finds simple prompting inadequate across linguistic, behavioral, and cognitive simulation criteria [34]. We therefore use classroom simulation as an observable testbed for interactive-agent strategy evaluation, treating action-pattern diversity, expert alignment, rationale–action association, and observable knowledge structures as distinct diagnostics. We do not treat generated classroom trajectories as evidence of authentic learner behavior, instructional appropriateness, or learning outcomes.

3 Method

LA-IQL models an interactive agent as a sequential decision maker: at each step, it observes dialogue history, chooses a domain-defined interaction action, and realizes that action in natural language. Authentic human trajectories ground the action space and supervision, while the explicit action boundary keeps strategy selection inspectable apart from

Category	Symbols	Role expansion
Prior-known knowledge	Ipk, Pk	student/teacher $\times$ symbol
Personal information	Ipi, Pi	student/teacher $\times$ symbol
Analysis	Ia, An	student/teacher $\times$ symbol
Coordination	Ic, Co	student/teacher $\times$ symbol
Speculation	Is, Sp	student/teacher $\times$ symbol
Uptake	Iu, Up	student/teacher $\times$ symbol
Agreement	Ag	student/teacher $\times$ symbol
Querying	Qu	student/teacher $\times$ symbol
Guide	Gu	student/teacher $\times$ symbol
END	episode termination	role-independent

Table 1. Compact definition of the 31-action interaction space.

language realization. Building on offline inverse reinforcement learning, LA-IQL remains compatible with standard supervised fine-tuning pipelines while applying a dynamics-aware TD regularizer to human demonstration trajectories. LA-IQL implements the action-mediated formulation through rationale-conditioned action prediction, offline sequential policy learning over human-readable acts, and separate language realization. The associated long-horizon evaluation examines alignment, recurrence, and context sensitivity as distinct properties of the resulting behavior. The learning objective is adapted from prior inverse soft Q-learning; the contribution lies in this formulation, its policy-learning implementation, and the diagnostic evaluation. We additionally examine proposal-constrained candidate re-ranking as an inference-time adjustment.

3.1 Problem Formulation and Interaction Action Space

We instantiate the interactive-agent problem as a finite-context Markov decision process (MDP) approximation for offline policy learning from expert classroom demonstrations. Let H_t denote the full interaction history before step t . The policy observation $s_t = f_{\text{short}}(H_t)$ consists of the previous two dialogue turns and two action labels; it is an approximate Markov state that may omit longer-range instructional context. The 15-symbol cognitive-act scheme and the utterance-level annotations are inherited from the prior classroom-analysis study [39]; we do not redefine the taxonomy or relabel the source turns in this work. The action a_t is the next interaction act to be generated. It is drawn from a 31-action discrete space: $\{\text{student, teacher}\} \times 15$ cognitive-act symbols (Prior-known knowledge, Personal information, Analysis, Coordination, Speculation, Uptake, Agreement, Querying, and Guide), plus a terminal symbol END. A selected utterance u_t and its action advance the history as $H_{t+1} = H_t \oplus (a_t, u_t)$, after which $s_{t+1} = f_{\text{short}}(H_{t+1})$; END marks episode termination. Thus the deployed transition is jointly induced by the action policy and the language realization model rather than by a_t alone. Given expert demonstrations $\mathcal{D}_E = \{(s_t, a_t, s_{t+1})\}$, we learn an action policy whose generated trajectories are evaluated for diversity and similarity to expert action patterns; language realization is evaluated separately. This action-level formulation is the first LA-IQL design choice: instead of applying inverse RL directly to open-ended tokens, we optimize over human-readable teacher–student cognitive acts and leave surface realization to a separate role model.

Table 2 summarizes the common implementation and rollout configuration.

Table 1 is the compact policy vocabulary; Appendix A.1 gives operational definitions and role-specific corpus examples for all 15 symbols. We use this taxonomy as an operational representation of interaction strategy, not as evidence that it is complete or universally valid across settings.

We evaluate diversity and expert-pattern alignment across systems (RQ1) and between checkpoints and inference settings (RQ2), and rationale–action consistency (RQ3). Because interaction actions determine whether an agent probes, elaborates, connects, or redirects information, their long-horizon organization may shape how information is introduced and related in generated dialogue. RQ4 therefore examines how differences in strategy selection are reflected in expressed knowledge structure, linking the action-level comparisons to the organization of dialogue content through evidence-grounded graph endpoints computed over matched long-horizon trajectories. Together, these goals operationalize the dimensions exposed by the interactive-agent framing: alignment, temporal organization, action-relevant reasoning, language realization, and long-horizon knowledge structure.

3.2 Data Construction and Rationale Annotation

Each expert classroom trajectory is converted into fixed-window transitions. For step t , s_t contains the two most recent utterances and the two most recent action labels, a_t is the annotated next interaction act, and s_{t+1} is obtained after appending the next utterance together with a_t . We use the deepseek-chat API (temperature 0.7) to rewrite rationale targets for 9,168 non-END transitions from the local state and expert next action. The remaining 201 transitions are END actions and do not require rationale targets. The artifact records the model alias and call date but not an immutable provider snapshot. Because the annotation model is conditioned on the expert next action, these rationales are constructed supervision targets rather than independently elicited explanations of expert reasoning. We use them as textual intermediates for action prediction, without assuming that they faithfully reconstruct expert reasoning or causally explain the learned policy’s decisions. The trained model does not receive the expert action as an inference-time input. It instead learns the factorization

$$p_{\theta}(z_t, a_t | s_t) = p_{\theta}(z_t | s_t) \pi_{\theta}(a_t | s_t, z_t). \quad (1)$$

Rationale targets that copy an action symbol or canonical label are filtered before training to reduce trivial label leakage. Automated generation checks cover length, uniqueness, and retry status; they do not constitute a semantic-quality audit. We use a fixed supervision schema:

<Rationale> explanation text </Rationale> <Action> role-symbol label </Action>.

This construction turns each non-END expert transition into paired supervision for rationale generation and action-level policy learning without a manually specified scalar reward. Because the annotation model sees the expert action whereas the deployed rationale generator does not, training mixes teacher-forced and model-generated rationales to reduce this distribution shift.

3.3 From MLE to Expert Distribution Matching

LA-IQL augments supervised fine-tuning of a rationale-conditioned dual-head model with offline IQLearn. Whereas MLE predicts the observed action locally, the IQLearn term couples the policy to successive state values without online interaction or a learned simulator.

3.4 TD-Regularized Offline IQLearn Objective

Building on offline inverse soft Q-learning for language imitation [43], we adapt its token-level formulation to learn a rationale-conditioned policy over 31 explicit interaction actions. Grounded in IQ-Learn [11], the resulting objective combines maximum-likelihood imitation with a temporal-difference regularizer. We define the augmented policy state

as $x_t = (s_t, z_t)$. For teacher-forced examples, $x_t^* = (s_t, z_t^*)$ and $x_{t+1}^* = (s_{t+1}, z_{t+1}^*)$ use the corresponding annotated rationale targets; model-generated-rationale examples replace these targets with samples from $p_\theta(z | s)$. At inference, the language head generates z_t from s_t before action scoring. The objective contains four terms. First, $\mathcal{L}_{\text{lm}}$ is the language-modeling loss over the LLM-annotated rationale and action prefix. Second, the policy head is supervised with action-level MLE:

$$\mathcal{L}_{\text{policy}} = -\log \pi_\theta(a_t | x_t). \quad (2)$$

Third, the LM head induces a soft action distribution $p_{\text{LM}}(\cdot | x_t)$ by scoring the fixed action-label templates after the rationale. We align the policy head with this soft distribution using a KL consistency loss:

$$\mathcal{L}_{\text{consistency}} = D_{\text{KL}}(p_{\text{LM}}(\cdot | x_t) \| \pi_\theta(\cdot | x_t)). \quad (3)$$

Finally, following the formulation of Wulfmeier et al., the policy head produces $Q_\theta(x_t, a)$ for every $a \in \mathcal{A}$. The corresponding soft value and policy are

$$V_\theta(x_t) = \log \sum_{a \in \mathcal{A}} \exp Q_\theta(x_t, a), \quad (4)$$

$$\log \pi_\theta(a | x_t) = Q_\theta(x_t, a) - V_\theta(x_t).$$

For an expert transition (x_t, a_t, x_{t+1}) , define the action-level temporal-difference residual:

$$\delta_t := V_\theta(x_t) + \log \pi_\theta(a_t | x_t) - \gamma V_\theta(x_{t+1}). \quad (5)$$

We set $V_\theta(x_{t+1}) = 0$ when $a_t = \text{END}$. Since $V_\theta + \log \pi_\theta = Q_\theta$, the corresponding implicit reward is $\delta_t = Q_\theta(x_t, a_t) - \gamma V_\theta(x_{t+1})$. Together, $-\log \pi_\theta(a_t | x_t) + \lambda_4 \delta_t^2$ is the action-level counterpart of the MLE-plus-TD objective derived by Wulfmeier et al. These terms are combined as

$$\mathcal{L} = \mathcal{L}_{\text{lm}} + \lambda_2 \mathcal{L}_{\text{policy}} + \lambda_3 \mathcal{L}_{\text{consistency}} + \lambda_4 \delta_t^2. \quad (6)$$

The coefficient λ_4 is zero in Stages 1–2. In Stage 3 it is set to 1 for Full and remains 0 for no-TD, recovering the non-TD objective when the term is disabled.

Training uses three successive two-epoch LoRA stages. Stage 1 emphasizes structural validity; Stage 2 jointly optimizes language and policy objectives; and Stage 3 continues joint optimization while increasing the proportion of self-generated rationales. The base model is Qwen/Qwen3.5-0.8B [33], the maximum sequence length is 512, and the physical batch size is 4 with one gradient-accumulation step (effective batch size 4). All three stages use training seed 42 and gradient clipping at 1.0. The evaluated Full and no-TD configurations are matched except for the Stage 3 TD-loss weight, which is 1 for Full and 0 for no-TD. Appendix B records the stage-specific values and the fixed boundary of the archived provenance.

3.5 Rationale-Conditioned Policy Parameterization

The model uses an explicit rationale-conditioned dual-head design, which is the third LA-IQL design choice. The language head (causal LM) generates a `<Rationale>` span before the `<Action>` label; the policy head applies cross-attention pooling over the rationale span and outputs one value per action to parameterize $Q(x, a)$ and $\pi(a | x)$. Decisions are conditioned on the generated rationale rather than a single pooled representation.

Concretely, given an input token sequence:

- The LM head produces token-level scores for $\mathcal{L}_{\text{lm}}$ and, by scoring action-prefix templates over the 31 actions, yields a soft LM action distribution.

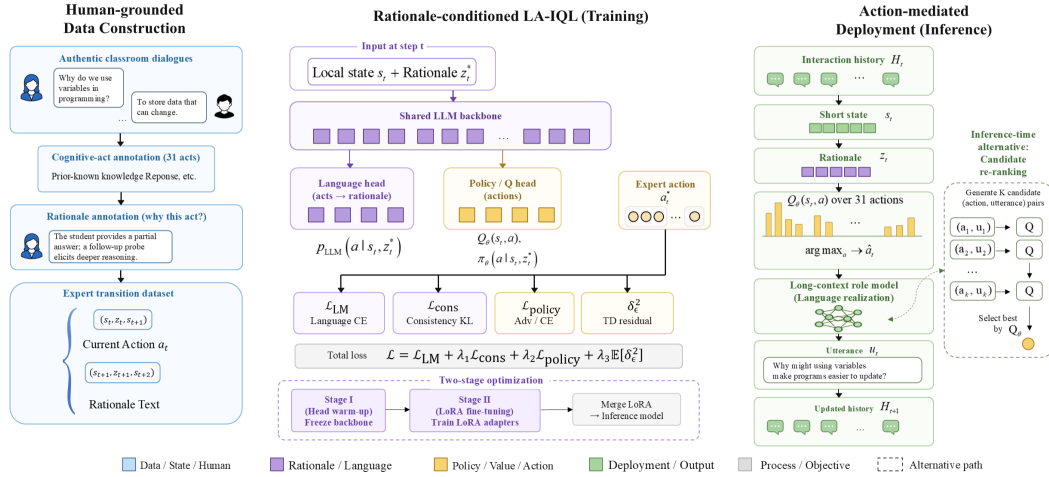


Fig. 3. **LA-IQL framework and technical flow.** Human-grounded classroom dialogue and cognitive-act annotations are converted into rationale-augmented expert transitions (left). Rationale-conditioned LA-IQL jointly trains language and policy/value components with language, consistency, policy, and temporal-difference objectives (center). At inference, the learned policy selects an interaction action and a long-context role model realizes it as an utterance; candidate re-ranking is shown as an inference-time alternative (right).

- The policy head pools the rationale text and outputs policy scores for $\mathcal{L}_{policy}$.
- A consistency loss $\mathcal{L}_{consistency}$ aligns the LM action-prefix distribution with the policy logits.

3.6 Direct Policy Decoding and Language Realization

At inference, LA-IQL separates short-context action selection from long-context language realization. Let $h_t = f_{long}(H_t)$ denote the role model's longer view of the same history, distinct from the policy observation s_t . The policy model first generates $z_t \sim p_\theta(z | s_t)$ and directly selects

$$\hat{a}_t = \arg \max_{a \in \mathcal{A}} Q_\theta(s_t, z_t, a). \quad (7)$$

The role model then produces one realization $u_t \sim q_\phi(u | h_t, \hat{a}_t)$. Thus the default LA-IQL inference rule exposes the full action space to the learned policy while using the longer context only to realize its selected action.

Figure 3 summarizes the corresponding data, optimization, and inference flow.

The role model is recorded as deepseek-chat for both teacher and student roles. It uses a long window of 30 dialogue turns and five action steps; where sampling applies, it uses a temperature of 0.8 and top- p of 1. The artifact retains the mutable API alias rather than an immutable model snapshot. In the reported evaluation, generation stops at END or the 100-action horizon.

We additionally explored candidate re-ranking as an inference-time adjustment. The role model samples six action-utterance candidates in a fixed schema—three for the teacher and three for the student—and thereby induces $q_\phi(\mathcal{C}_t | h_t)$

over $\mathcal{C}_t = \{(a_{t,j}, u_{t,j})\}_{j=1}^6$. The selected index is

$$j^* = \arg \max_{1 \leq j \leq 6} Q_\theta(s_t, z_t, a_{t,j}), \quad (8)$$

$$(a_t, u_t) = (a_{t,j^*}, u_{t,j^*}).$$

This adjustment scores a candidate through its associated action label and does not directly score $u_{t,j}$; candidates sharing an action are therefore indistinguishable to Q_θ , and ties retain proposal order. It is proposal-constrained because an action cannot be selected unless it appears in $\mathcal{C}_t$. Unlike default direct decoding, the role model therefore controls action support while the value model only chooses among proposed actions.

4 Experimental Design

Research questions. We answer four research questions. Together, they test whether an interactive agent’s explicit strategy layer is human-pattern-aligned, temporally organized, context-sensitive, and legibly realized:

RQ1: How do action-mediated learned policies compare with direct language generation and statistical policies in behavioral diversity, repetition, and alignment with held-out patterns from authentic human interaction?

RQ2: What long-horizon behavioral differences are observed between the evaluated fixed checkpoints with and without temporal-difference regularization, and between direct full-action-space selection and proposal-constrained candidate re-ranking under the completed configurations?

RQ3: Do generated rationales contain action-relevant information beyond the dialogue state and explicit label leakage?

RQ4: How are differences in interaction-strategy selection reflected in the introduction, connection, and development of expressed knowledge over long-horizon dialogue?

These questions primarily target model- and trajectory-level structural properties. Participant judgments are not used as evidence of educational effectiveness; we add a supplementary excerpt-level evaluation to examine whether differences between LA-IQL and direct language generation are perceptible in interactional variety, resemblance to teacher–student interaction, and support for understanding and discussion.

4.1 Data and Splits

The source classroom recordings were obtained from China’s National Public Service Platform for Educational Resources¹, a national platform supported by the Ministry of Education. The platform hosts K–12 lessons contributed voluntarily by teachers using classroom recording systems and covers schools across major provinces and municipalities in China. We selected lessons from this source because of its broad geographic and subject coverage, institutional provenance, and publicly accessible materials, which facilitate inspection of the source lessons. From the resulting classroom corpus described in Section 3.1, we use 50 lessons in this study. Each source turn retains its previously assigned cognitive-act label and recorded speaker role. We pair these two fields and add the role-independent END action to construct the 31 interaction acts and transition tuples used by LA-IQL. We use a frozen, stratified lesson-level split of 35 training, five validation, and 10 test lessons rather than splitting by dialogue turn, so that windows from the same classroom session cannot occur in both partitions. The split is stratified by subject and school stage; lesson identifiers are disjoint across all three partitions, and two lessons used in pilot smoke tests are excluded from the test partition.

¹<https://www.eduyun.cn>

Component	Recorded configuration
Policy model	Qwen/Qwen3.5-0.8B; 31-action rationale-conditioned dual head
Rationale annotation	deepseek-chat; temperature 0.7; 9,168 rewritten non-END records; 201 END transitions omitted
Role realization	deepseek-chat; temperature 0.8; top- p 1; 30-turn/5-action context
Policy observation	Two dialogue turns and two action labels; max input length 512; rationale budget 256 tokens
Formal split	35/5/10 lessons; subject-stage stratification; split seed 20260731
Rollout	10 test lessons; horizon 100; END disabled before step 50; margin 3.0
Randomness	Full/no-TD: one checkpoint each, generation seeds 42–44; BC: independent train/gen. seeds 42–44

Table 2. Implementation summary. Values are transcribed from checkpoint, split, annotation, and formal-trajectory artifacts. Historical training fields not retained by those artifacts are documented as unavailable in Appendix B.

The split manifest, random seed, data hash, and split-level action frequencies are recorded before formal evaluation. The same test lessons were inspected in an earlier horizon-20 run; we disclose this prior access and do not alter test membership for the horizon-100 baseline evaluation reported below.

4.2 Baselines, Controls, and System Comparisons

Our comparisons have four distinct roles and should not be read as one set of exchangeable baselines.

Generation baselines. Direct role-conditioned generation and a constrained teacher prompt test what end-to-end prompted generation produces without a separately learned action policy. They provide output-level references, not component-matched ablations of LA-IQL.

Learned-policy comparisons. Frozen BC and LoRA BC are supervised next-action predictors trained on the same 31-action space and source split. Frozen BC trains only a 31-way classification head over a frozen Qwen/Qwen3.5-0.8B backbone; LoRA BC adapts the same backbone with LoRA under the same next-action cross-entropy objective. Neither uses rationales, a TD objective, inverse Q-learning, or Q-value decoding. Both use three independently trained checkpoints and realize selected actions with the same role model and formal horizon-100 rollout protocol. They test how the implemented one-step supervised policy family behaves under the shared action interface; because several training components differ together, BC–LA-IQL differences do not isolate any single LA-IQL component.

Statistical diagnostic controls. Uniform action sampling, expert-frequency sampling, and a first-order Markov policy isolate distributional and sequential statistics. They are not learned, state-conditioned interactive policies and are not candidates for overall system superiority. In particular, the Markov control estimates $p(a_t \mid a_{t-1})$ from the frozen training split. It is deliberately well matched to bigram- and transition-based alignment measures, but it does not observe dialogue content, the current instructional state, or a rationale, and it does not produce an utterance whose realization can be inspected. Its role is therefore to test how much local human-pattern alignment can be recovered from aggregate action transitions alone.

Inference-time system comparison. LA-IQL uses the direct decoding rule in Equation 7. *LA-IQL + candidate re-ranking* uses the same learned policy checkpoint but substitutes the proposal-constrained rule in Equation 8; it is therefore an inference-time system configuration, not a separately trained baseline. This comparison was elevated after inspection of the current held-out results and is descriptive rather than a preregistered model-selection claim. Both configurations

use the same checkpoint, lessons, horizon, role-model temperature, and generation seeds. However, Direct uses rationale temperature 0.0 whereas Candidate uses 0.6, so their differences characterize the completed systems and cannot be attributed solely to candidate re-ranking.

TD checkpoint contrast. To examine the checkpoint-level difference associated with temporal-difference regularization, we add *LA-IQL w/o TD (Direct)*, trained with Stage 3 TD weight 0 rather than 1. It uses Direct decoding, rationale temperature 0.0, the same held-out lessons, role model, context windows, generation seeds, horizon, and stopping rule as the main LA-IQL system. The recovered training fields otherwise match (Appendix B), but each condition contains one trained checkpoint; we therefore treat this as a fixed-checkpoint contrast rather than a training-seed-general causal estimate.

4.3 Evaluation Metrics

For interaction-pattern diversity, the main analysis reports action Distinct-2, two-step cycle rate, empirical action entropy, and expert-pattern coverage, capturing local variety, repetitive degeneration, distributional diversity, and expert-supported breadth, respectively. For expert-pattern alignment, we use Jensen–Shannon divergence between generated and expert action patterns, transition-matrix distance, and expert support for generated action n -grams. These are behavioral-distribution measures, not tests of whether an action is contextually appropriate for the dialogue state. Their overlap with the Markov control’s inductive bias makes that control a demanding local-pattern reference rather than evidence of state-sensitive interaction competence. For rationale–action consistency (RQ3), we mask deterministic label strings and train an independent 31-way action predictor on disjoint lessons using the state, the rationale, or both. Our principal automatic measure is incremental simulatability:

$$\Delta_{\text{sim}} = F_1^{\text{macro}}(s_t, z_t \rightarrow a_t) - F_1^{\text{macro}}(s_t \rightarrow a_t), \quad (9)$$

where the target is the evaluated system’s selected action. We also report masked rationale-only Macro-F1, majority, leakage, and shuffled-rationale controls. Predictability alone does not establish causal faithfulness [12, 16]; we therefore describe RQ3 as a simulatability diagnostic rather than an explanation-quality measure. For secondary language-realization diagnostics, we report redundancy and rule-defined severe violations. This deliberately narrow diagnostic set does not measure contextual or classroom appropriateness; open-ended text overlap is secondary because multiple responses can be valid. For RQ4, each system–generation-seed–lesson trajectory is a measurement unit. Repeated runs are averaged within each system–lesson unit before paired comparisons; the held-out lesson is the unit of statistical inference and cluster resampling. Its primary evidence is an evidence-grounded knowledge-graph extraction followed by deterministic calculation of three prespecified endpoints: unique entities per 100 non-END turns, mean relational degree, and the proportion of final relations first introduced in the latter half of a trajectory. Rule-based diagnostics provide auditable anchors. We report the knowledge-graph endpoints without combining them into a classroom-quality score. Appendix D gives the complete metric and intervention protocol.

4.4 Evaluation Protocol

Action-pattern Jensen–Shannon divergence is the primary alignment endpoint. Diversity and alignment jointly address cross-system comparisons (RQ1) and checkpoint and inference-configuration comparisons (RQ2). The reported trajectory results cover seeds 42–44 and 10 frozen test lessons per seed at horizon 100. For the LA-IQL configurations and generation controls, these are three stochastic rollouts from a fixed checkpoint or inference procedure. For Frozen

BC and LoRA BC, each seed identifies an independently trained checkpoint and its matching generation run. We report mean $\pm$ population standard deviation across seed aggregates; the BC variation thus combines training and generation variation, whereas the LA-IQL variation does not include training-seed uncertainty. For learned-system comparisons, we additionally pair the 30 matching generation-seed-lesson trajectories, cluster resampling by the 10 lessons, run 10,000 bootstrap resamples, and report percentile 95% intervals. The reported TD p -values are unadjusted within an exploratory matched-checkpoint family; the Direct-Candidate table applies Benjamini-Hochberg correction over its 10 retained comparisons. The TD contrast holds Direct decoding and rationale temperature fixed. By contrast, Direct and Candidate differ in rationale temperature and were examined after observing held-out results, so that comparison remains a descriptive system-level contrast rather than a confirmatory causal decoding effect.

Supplementary human evaluation. Thirty-five adult volunteers completed the shortened questionnaire; two records contained no excerpt ratings, leaving 33 complete responses (25 women and eight men). Each participant independently rated six blinded, same-topic excerpt sets on three five-point items: interactional variety, resemblance to authentic teacher-student interaction, and support for understanding and discussion. For the reported model comparison, we aggregate the six LA-IQL Direct and six Direct-role LLM ratings within participant and dimension, apply paired Wilcoxon signed-rank tests, and use Holm correction across the three dimensions. Because removing four sets from the longer instrument left source positions unbalanced, we additionally fit a sensitivity model with participant, item, dimension, and display-position effects and participant-clustered standard errors. The questionnaire also collected ratings for authentic transcript excerpts, but unavoidable transcription loss and differences in excerpt length and exact dialogue state made those scores non-equivalent to the generated excerpts. We therefore exclude that reference condition from model-comparison inference and treat this evaluation as supplementary and exploratory.

Table 3 reports cross-system diversity results (RQ1).

Table 4 reports cross-system alignment results (RQ1) on the same runs. Paired JS uses the frozen 31^2 bigram vocabulary and Jeffreys smoothing ($\alpha = 0.5$); transition JS weights rows by expert frequency, and support uses training-split n -grams.

4.5 Experimental Results

The evaluated TD checkpoint shows less repetitive long-horizon organization. Table 5 reports the matched Direct-decoding checkpoint contrast. The w/o-TD checkpoint has more two-step cycles (0.2607 vs. 0.1708; TD-w/o-TD difference -0.0900 , 95% CI $[-0.1400, -0.0438]$, $p = .0002$) and greater motif recurrence (0.5379 vs. 0.3957; difference -0.1422 , 95% CI $[-0.2013, -0.0889]$, $p = .0002$). Relative to the TD-regularized checkpoint, these correspond to 53% more two-step cycles and 36% more repeated motifs for the w/o-TD checkpoint. Distinct-2 and normalized entropy are numerically higher with TD but do not differ reliably. Both systems cover all top-20 expert patterns, supporting the interpretation that, for these checkpoints, lower recurrence does not coincide with abandoning those common patterns.

Local expert-distribution alignment changes little. Paired bigram JSD is 0.0827 with TD and 0.0835 without TD (difference -0.0007 , 95% CI $[-0.0063, 0.0043]$, $p = .8281$); transition JSD is likewise not reliably different. Bigram support shows a borderline descriptive advantage for TD (0.9906 vs. 0.9833, $p = .0510$). For these fixed checkpoints, the observed TD-associated difference is concentrated in recurrent sequence structure rather than local imitation fidelity. Our inference is therefore explicitly bounded to the evaluated checkpoints rather than the training objective across random initializations.

System or control	Dist.-2 $\uparrow$	Cycle-2 $\downarrow$	H_{norm} $\uparrow$	Cov.@20 $\uparrow$
<i>Generation baselines</i>				
Direct Role LLM	0.4364 ± 0.0348	0.0286 ± 0.0088	0.6722 ± 0.0102	0.5500 ± 0.0408
Constrained Prompt	0.5279 ± 0.0117	0.1520 ± 0.0147	0.6818 ± 0.0015	0.5333 ± 0.0236
<i>Statistical diagnostic controls</i>				
Uniform Random	0.9593 ± 0.0048	0.0000 ± 0.0000	0.9287 ± 0.0066	0.3500 ± 0.0000
Expert Frequency	0.6951 ± 0.0277	0.0067 ± 0.0033	0.7144 ± 0.0111	0.9500 ± 0.0000
First-order Markov	0.5144 ± 0.0072	0.0819 ± 0.0140	0.7072 ± 0.0012	1.0000 ± 0.0000
<i>Learned-policy comparisons</i>				
Frozen BC	0.0694 ± 0.0062	0.0072 ± 0.0069	0.2379 ± 0.0134	0.3167 ± 0.1434
LoRA BC	0.0539 ± 0.0267	0.0017 ± 0.0013	0.1267 ± 0.0940	0.3333 ± 0.2055
<i>LA-IQL checkpoint and inference configurations</i>				
LA-IQL w/o TD (Direct)	0.4118 ± 0.0236	0.2607 ± 0.0193	0.6412 ± 0.0120	1.0000 ± 0.0000
LA-IQL	0.4370 ± 0.0101	0.1708 ± 0.0126	0.6670 ± 0.0084	1.0000 ± 0.0000
LA-IQL + candidate re-ranking	0.4478 ± 0.0205	0.0488 ± 0.0013	0.6664 ± 0.0131	0.7000 ± 0.0408

Table 3. Cross-system diversity results for RQ1 (mean $\pm$ population SD over three completed runs). Row groups denote different evidential roles rather than a single fairness-matched ranking. Bold values are restricted to the learned-policy and LA-IQL configuration rows and do not imply cross-category system superiority. Metrics must be interpreted jointly because low Cycle-2 alone does not exclude other forms of recurrence.

System or control	Paired bigram JS $\downarrow$	Pooled bigram JS $\downarrow$	Transition JS $\downarrow$	Bigram support $\uparrow$	Trigram support $\uparrow$
<i>Generation baselines</i>					
Direct Role LLM	0.1030 ± 0.0013	0.3423 ± 0.0184	0.1708 ± 0.0049	0.7532 ± 0.0167	0.4250 ± 0.0267
Constrained Prompt	0.1096 ± 0.0016	0.3795 ± 0.0072	0.1880 ± 0.0069	0.9218 ± 0.0092	0.4731 ± 0.0144
<i>Statistical diagnostic controls</i>					
Uniform Random	0.1235 ± 0.0012	0.4661 ± 0.0042	0.2293 ± 0.0028	0.4105 ± 0.0249	0.0431 ± 0.0047
Expert Frequency	0.1036 ± 0.0006	0.2868 ± 0.0036	0.1995 ± 0.0036	0.8125 ± 0.0161	0.3316 ± 0.0032
First-order Markov	0.0791 ± 0.0009	0.0982 ± 0.0038	0.1249 ± 0.0017	0.9935 ± 0.0021	0.8557 ± 0.0169
<i>Learned-policy comparisons</i>					
Frozen BC	0.1607 ± 0.0075	0.5807 ± 0.0347	0.3053 ± 0.0166	0.9983 ± 0.0017	0.7412 ± 0.2445
LoRA BC	0.1674 ± 0.0057	0.6041 ± 0.0309	0.2811 ± 0.0255	0.9949 ± 0.0071	0.9143 ± 0.0847
<i>LA-IQL checkpoint and inference configurations</i>					
LA-IQL w/o TD (Direct)	0.0835 ± 0.0022	0.1530 ± 0.0177	0.1308 ± 0.0041	0.9833 ± 0.0005	0.8773 ± 0.0125
LA-IQL	0.0827 ± 0.0031	0.1637 ± 0.0085	0.1299 ± 0.0050	0.9906 ± 0.0024	0.8772 ± 0.0064
LA-IQL + candidate re-ranking	0.1059 ± 0.0025	0.3116 ± 0.0132	0.1648 ± 0.0035	0.9522 ± 0.0045	0.4874 ± 0.0063

Table 4. Cross-system expert-pattern alignment for RQ1 (mean $\pm$ population SD over three completed runs). Row groups have the distinct evidential roles defined in Section 4.2; bold is restricted to learned-policy and LA-IQL configuration rows. The first-order Markov control directly models the local transitions measured by several columns but does not condition on dialogue content.

The implemented one-step BC policies exhibit narrow recurrent patterns. The two supervised BC systems provide learned-policy references rather than single-component ablations. In the completed runs, Frozen BC and LoRA BC obtain Distinct-2 values of 0.0694 and 0.0539, normalized entropies of 0.2379 and 0.1267, and repeated-motif rates of 0.9690 and 0.9697, respectively. Direct LA-IQL is substantially more diverse and less motif-recurrent on the same

Metric family	Paired metric (TD – w/o TD)	Difference	95% CI	p
Diversity	Distinct-2 $\uparrow$	+0.0252	[−0.0282, 0.0688]	0.3186
Diversity	Cycle-2 $\downarrow$	−0.0900	[−0.1400, −0.0438]	0.0002
Diversity	Motif rate $\downarrow$	−0.1422	[−0.2013, −0.0889]	0.0002
Diversity	Normalized entropy $\uparrow$	+0.0258	[−0.0156, 0.0659]	0.2204
Alignment	Paired bigram JSD $\downarrow$	−0.0007	[−0.0063, 0.0043]	0.8281
Alignment	Transition JSD $\downarrow$	−0.0009	[−0.0091, 0.0055]	0.8581
Alignment	Bigram support $\uparrow$	+0.0073	[−0.00001, 0.0140]	0.0510
Secondary	Local trigram redundancy $\downarrow$	−0.00001	[−0.0042, 0.0041]	—

Table 5. Fixed-checkpoint temporal-difference contrast under Direct decoding. Values compare LA-IQL with and without the TD term over the same 30 generation-seed-lesson trajectories, clustered by 10 held-out lessons, with 10,000 bootstrap resamples. Both systems use rationale temperature 0.0 and otherwise identical generation settings. Positive differences mean the TD model is numerically larger; metric arrows indicate the desirable direction.

endpoints (Distinct-2 0.4370, normalized entropy 0.6670, and motif rate 0.3957). The BC Cycle-2 rates are low, but this does not indicate broad temporal organization: their near-unit motif rates show that they predominantly repeat a narrow set of local patterns rather than alternate in the specific two-step form measured by Cycle-2. The complete recurrence results appear in Table 13.

The same baselines also reproduce training-supported local transitions without matching the held-out distribution closely. Frozen BC and LoRA BC reach bigram support of 0.9983 and 0.9949, but their paired bigram JSD values are 0.1607 and 0.1674, compared with 0.0827 for Direct LA-IQL. LoRA BC has higher trigram support (0.9143 vs. 0.8772), illustrating why support must be read jointly with distributional divergence and recurrence. These descriptive comparisons show that, for these implemented BC configurations, high local support does not coincide with diverse or expert-distribution-aligned long-horizon organization. They do not isolate rationales, TD regularization, or Q-value decoding as the sole cause of the difference.

Local transition statistics form a strong but limited reference. Tables 3 and 4 report the formal trajectory measurements. The first-order Markov control obtains the lowest paired bigram JSD (0.0791), pooled bigram JSD (0.0982), and transition-row JSD (0.1249), compared with 0.0827, 0.1637, and 0.1299 for Direct LA-IQL. It also reaches 0.9935 bigram support, while Direct LA-IQL is higher on trigram support (0.8772 vs. 0.8557).

This result follows from a substantive match between model and measurement: the control estimates $p(a_t | a_{t-1})$ from training trajectories, while bigram JSD and transition-row JSD measure the corresponding local action structure. High support and coverage can likewise arise from conservatively resampling training-supported transitions. These are legitimate strengths on the stated alignment endpoints, not an error to discount. However, the control never observes the dialogue text or instructional state and cannot expose whether a locally frequent transition is appropriate in context. We therefore interpret it as evidence that local human-pattern reproduction is achievable with a simple sequential statistic, and not as evidence that the Markov policy is a more capable interactive agent. This distinction motivates reporting alignment alongside state-conditioned decision traces, repetition diagnostics, and action-utterance realization checks.

A frozen intervention provides exploratory evidence of text sensitivity. We evaluated the main Full checkpoint on five pre-existing, evenly spaced states from each of the 10 held-out lessons ($N = 50$). Within each lesson, a cyclic donor replaced utterance semantics while preserving the speaker sequence, number of utterances, and action history.

Metric family	Paired metric (Direct – Candidate)	Difference	95% CI	BH q
Diversity	Distinct-2 $\uparrow$	−0.0108	[−0.0377, 0.0165]	0.5483
Diversity	Cycle-2 $\downarrow$	+0.1220	[0.0986, 0.1447]	< .0001
Diversity	Motif rate $\downarrow$	+0.2543	[0.2100, 0.2963]	< .0001
Alignment	Paired bigram JSD $\downarrow$	−0.0232	[−0.0305, −0.0152]	< .0001
Alignment	Bigram support $\uparrow$	+0.0384	[0.0283, 0.0492]	< .0001
Alignment	Trigram support $\uparrow$	+0.3898	[0.3551, 0.4238]	< .0001

Table 6. Paired Direct–Candidate comparison over 30 matched generation–seed–lesson units (10,000 bootstrap resamples; percentile intervals; BH correction over 10 retained comparisons). Positive values mean Direct is numerically larger; metric arrows indicate the desirable direction. The configurations also differ in rationale temperature, so this is a system-level comparison rather than a single-factor decoding ablation.

Relative to the original states, the probability assigned to the expert action decreased by 0.0767 (lesson-cluster bootstrap 95% CI [0.0153, 0.1288]), 60% of argmax actions flipped (CI [0.48, 0.72]), and mean policy JSD was 0.2957 (CI [0.2476, 0.3431]). The 31-class Macro-F1 decrease was 0.0201, with an interval crossing zero ([−0.0185, 0.0649]). This intervention indicates that the fixed policy distribution responds to dialogue text beyond the preserved action history. Its small sample, deterministic donor construction, and lack of human judgments mean it does not establish that the response is contextually appropriate. Appendix C records the protocol.

Candidate re-ranking changes the inference-time control trade-off. Direct LA-IQL is closer to the held-out expert distribution than Candidate: paired bigram JSD is 0.0827 versus 0.1059, and training-expert bigram support is 0.9906 versus 0.9522. Across 30 matched generation–seed–lesson units, the Direct–Candidate difference in bigram JSD is −0.0232, 95% CI [−0.0305, −0.0152], BH-adjusted $q < .0001$.

This alignment does not imply uniformly better interaction dynamics. Direct has substantially more two-step cycles (0.1708 vs. 0.0488; paired difference +0.1220, 95% CI [0.0986, 0.1447], $q < .0001$) and repeated motifs (0.3957 vs. 0.1413; difference +0.2543, 95% CI [0.2100, 0.2963], $q < .0001$). Distinct-2 has no reliable paired difference (−0.0108, 95% CI [−0.0377, 0.0165], $q = .5483$). The completed configurations therefore differ more clearly in action-sequence organization than in local combinatorial diversity. Because Candidate reuses the Direct policy checkpoint and also changes rationale temperature, this is a comparison of two inference-time systems, not evidence that a separately learned Candidate policy is superior or an isolated ablation of re-ranking.

A rule-selected trace makes the recurrence difference inspectable. Before reading trajectory text, we selected the matched lesson–seed unit with the largest Direct–Candidate Cycle-2 difference, breaking any tie by motif-rate difference and then lexical order. This rule selects the Grade-9 chemistry lesson on atomic structure at generation seed 43 (Cycle-2: 0.309 vs. 0.052; motif rate: 0.55 vs. 0.17). Table 7 shows the earliest four-turn Direct cycle and the same turn positions in the matched Candidate trajectory. Rationales and utterances are abridged translations from Chinese; action symbols follow Table 1.

The action trace exposes recurring decision types that would be difficult to audit from surface text alone. At the same time, the Direct utterances advance from a claim about nuclear force to the countervailing role of electrostatic repulsion. The Candidate trace also exposes a local rationale–realization shift at t15: the recorded rationale focuses on refining placement and mass, while the utterance asks about electrostatic stability. The example therefore illustrates, but does not by itself validate, the aggregate recurrence contrast or realization quality: a repeated action motif is not

Configuration and local context	Four-turn decision trace: action; rationale → utterance
Direct	
Student had just restated the teacher's explanation of nuclear force.	t13 Teacher-Ag: confirm understanding and extend the issue → "Exactly. Does having more protons mean stronger nuclear force and a more stable nucleus?" t14 Student-Pk: answer the hypothesis from prior knowledge → "I think so; more protons should mean more nuclear force." t15 Teacher-Ag: affirm the analogy, then introduce a competing force → "But electrostatic repulsion also increases; which force wins?" t16 Student-Pk: respond to that comparison → "Repulsion also grows, so the nucleus may not necessarily become more stable."
Candidate	
Student had just estimated electron mass and located most atomic mass in the nucleus.	t13 Teacher-Gu: confirm the estimate and consolidate mass-distribution understanding → "Draw an atomic model and mark each part's position and approximate mass share." t14 Student-An: construct and explain the requested model → "A small central nucleus contains protons and neutrons; light electrons move outside it." t15 Teacher-Iu: refine the diagram's component placement and mass analogy → "What force keeps the electrons and nucleus in a stable relation?" t16 Student-Pk: supply the requested relation → "The attraction between negative electrons and the positive nucleus."

Table 7. Rule-selected inspectable case from formal trajectories (lesson 4873, generation seed 43, turns 13–16). The Direct window instantiates an Ag-Pk-Ag-Pk two-step cycle, whereas Candidate traverses Gu-An-Iu-Pk. The entries preserve the recorded action, rationale intent, and utterance content while abbreviating the original text for space.

equivalent to verbatim repetition or instructional failure. We consequently use the case for structural interpretation rather than a human-preference ranking.

Rationales carry action-predictive information. Under the same RQ3 dual TF-IDF evaluator, masked rationale-only Macro-F1 is 0.3986 for Direct and 0.1013 for Candidate, compared with state-only scores of 0.0159 and 0.0123. Run-level incremental simulatability Δ_{sim} is 0.3885 for Direct and 0.0863 for Candidate. The w/o-TD Direct checkpoint obtains rationale-only Macro-F1 0.3682, combined Macro-F1 0.3629, $\Delta_{\text{sim}} = 0.3372$, and accuracy 0.6867, descriptively below the corresponding TD Direct values of 0.3986, 0.4044, 0.3885, and 0.7097. For the original Direct and Candidate configurations, the observed run-level delta exceeds all 20 within-lesson/role shuffle controls, and all 30 run-lesson deltas are positive. These results show that masked rationales contain information about the selected action. They do not show that rationales causally determine the policy decision or constitute faithful explanations; we also do not attach an inferential TD-effect claim to the descriptive RQ3 difference.

Rule-based diagnostics delimit language-realization risk. The final language-realization evidence consists of conservative, reproducible rule-based checks. They show no exact repetitions or rule-defined severe violations for Direct, Candidate, or w/o-TD Direct, while local trigram redundancy does not differ reliably between Direct and Candidate or between the two Direct checkpoints. These narrow checks do not measure realization quality or classroom appropriateness.

Supplementary ratings favor LA-IQL on variety and discussion support. Figure 4 reports the two-system comparison from 33 complete responses. LA-IQL receives higher ratings than Direct-role LLM for interactional variety (4.09 vs. 3.61; paired difference 0.48, Holm-adjusted $p < .001$) and support for understanding and discussion (4.23 vs. 3.77; difference 0.46, adjusted $p < .001$). Ratings of resemblance to authentic teacher-student interaction are comparable (3.91 vs. 3.79; difference 0.13, adjusted $p = .299$). After accounting for the unbalanced A/B/C positions, the estimated LA-IQL advantages remain 0.52 for variety and 0.45 for discussion support, whereas the adjusted resemblance difference is 0.07. The cross-system action-pattern results (RQ1) support closer alignment with held-out human action patterns than the

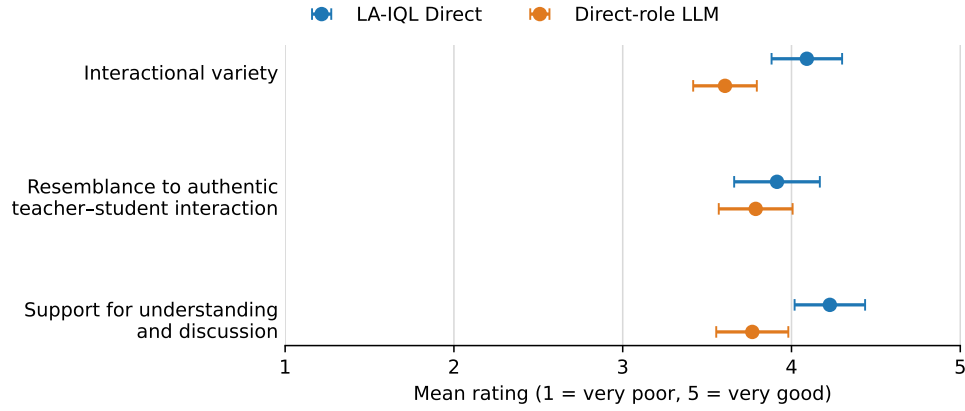


Fig. 4. Supplementary excerpt-level human evaluation. Points are means of participant-level ratings across six excerpts per system and dimension; error bars are 95% confidence intervals. The figure contains only the main LA-IQL Direct system and the Direct-role LLM generation baseline. The first and third dimensions differ under Holm-corrected paired tests ($p < .001$), whereas perceived resemblance does not ($p = .299$).

System	UER ₁₀₀	Mean relational degree	Late-stage relation growth
Expert transcript	14.17	1.906	0.456
LA-IQL Direct	23.95	2.112	0.379
LA-IQL Candidate	31.14	2.116	0.379
LA-IQL w/o TD (Direct)	27.96	1.963	0.380
Direct-role LLM	38.65	2.159	0.334
Constrained teacher prompt	39.15	2.164	0.360
Frozen BC	30.36	2.012	0.301
LoRA BC	32.22	2.064	0.325
Likelihood rerank	28.32	2.202	0.335

Table 8. Knowledge-graph endpoints for RQ4. UER₁₀₀ is the number of evidence-grounded unique entities per 100 non-END turns. Generated-system values first average the three completed runs within each of the 10 held-out lessons. The expert row contains one transcript per lesson. These endpoints describe generated knowledge structure, not factuality or instructional quality.

Direct-role LLM baseline. The supplementary ratings support higher perceived variety and discussion support; no clear difference in perceived classroom-interaction resemblance was detected.

Strategy-selection configurations also differ in expressed knowledge structure. RQ4 extends the action-level comparisons to the content expressed in generated dialogue, examining whether strategy-selection configurations are accompanied by differences in knowledge introduction, connection, and development. All 250 extracted graphs passed the machine-checkable schema and utterance-ID evidence constraints. Table 8 reports the three prespecified endpoints after first averaging repeated runs within each system-lesson unit. In addition to the systems used in the main RQ1/RQ2 comparisons, RQ4 includes completed Frozen BC and LoRA BC trajectories and an LM-likelihood reranking baseline. Full implementation details are reported in Appendix B.

Relative to Candidate, Direct introduces 7.19 fewer unique entities per 100 turns (95% CI $[-10.43, -4.25]$, Holm-adjusted $p = .0059$), while mean relational degree (-0.003 , CI $[-0.098, 0.093]$) and late-stage relation growth (-0.001 ,

CI $[-0.065, 0.052]$) are nearly unchanged. Relative to the matched w/o-TD Direct checkpoint, Direct is lower in UER₁₀₀ by 4.00 (CI $[-8.69, -0.05]$, adjusted $p = .2109$), higher in mean relational degree by 0.149 (CI $[0.059, 0.240]$, adjusted $p = .0527$), and indistinguishable in late-stage growth (-0.001 , CI $[-0.048, 0.050]$). These results support the interpretation that differences between the evaluated strategy-selection configurations are accompanied by differences in expressed content organization: Direct introduces fewer distinct entities than Candidate, with similar observed mean relational degree and late-stage relation growth. The latter similarities are descriptive and do not establish equivalence. This extends the comparison beyond action labels, while the system-level contrast does not isolate strategy selection as the cause of these differences. UER₁₀₀ has no intrinsic “higher is better” direction, so this result should not be interpreted as an educational-quality ranking.

Appendix A gives the full system-level comparison, RQ3 table, and judgment status.

5 Discussion

5.1 Human Grounding as an Inspectable Design Resource

Our first contribution is methodological: action mediation renders human grounding an operational and inspectable design resource for building more natural interactive agents. The action vocabulary, training demonstrations, and held-out comparison patterns are all drawn from authentic classroom interaction. It links policy learning, generation, and evaluation through a shared human-readable strategy layer. This goes beyond structured prompting for next-utterance generation: it establishes a stable interface at which designers can inspect what strategy was selected, how strategies unfold over time, and how each choice is realized in language. For HCI practitioners seeking to move beyond agents that merely answer questions toward agents that guide, probe, and collaborate, such traceability is essential. This is because it distinguishes between hoping a model has acquired sound interactional habits and verifying what it has actually learned.

The metric-matched Markov control clarifies what human-pattern alignment can and cannot establish. Its strong local alignment indicates that transition similarity can be reproduced without reading the dialogue; the semantic intervention, by contrast, shows that the learned policy distribution responds to dialogue content beyond the preserved action history. Human-pattern similarity is therefore informative only when situated within a broader, inspectable evaluation rather than treated as a sufficient measure of context-sensitive interaction. This finding carries a cautionary implication for the broader enterprise of learning from human demonstrations: matching surface statistics of human behavior does not guarantee that the agent has internalized the underlying interactional competence. Human grounding still does not ensure globally human-like or contextually appropriate behavior, but the action boundary makes the relevant successes and failures empirically separable, and therefore addressable.

5.2 TD-Associated Differences Support a Longer-Horizon Interpretation

The matched Direct-decoding checkpoint contrast isolates the clearest observed TD-associated difference. The w/o-TD checkpoint shows no reliably detected difference in local expert-pattern alignment yet exhibits 53% more two-step cycles and 36% more repeated motifs. This pattern is consistent with the TD term affecting recurrent sequence organization rather than merely expanding the action vocabulary or improving one-step imitation, though the single checkpoint per condition precludes a training-objective-level causal conclusion. The w/o-TD policy also terminates five of 30 trajectories before the 100-action horizon, whereas the TD policy terminates none, providing further descriptive

evidence of divergent rollout behavior. Because each condition contains one trained checkpoint, the bootstrap captures held-out lesson and generation variation but not training-seed uncertainty.

This fixed-checkpoint result supports the analytical value of the action-mediated formulation for identifying differences in recurrent interaction organization. It localizes the observed TD-associated difference in temporal organization (i.e., 53% more two-step cycles and 36% more repeated motifs for the evaluated w/o-TD checkpoint) rather than in local expert-pattern alignment or action-vocabulary expansion. This distinction bears directly on user experience: an agent that cycles through a small repertoire of interaction moves, however locally appropriate, will feel monotonous and predictable in sustained use. End-to-end text evaluation would obscure this distinction, since surface-level lexical variation can mask underlying interactional rigidity. The supplementary excerpt ratings indicate perceived gains in interactional variety and support for discussion, but they do not establish sustained coherence, experienced usefulness, or educational benefit. Understanding how the structural properties translate into extended user experience remains future work.

5.3 Action Mediation Makes Control Allocation Observable

Separating strategy selection from language realization turns control allocation into an observable design variable. The completed Direct configuration grants the learned policy access to the full action space and yields stronger alignment with held-out human action patterns, but also produces more short cycles and repeated motifs. The Candidate configuration restricts action support to role-model proposals and exhibits fewer repetitions, yet its trajectories diverge further from the held-out action distribution. Because the configurations also differ in rationale temperature, this constitutes a descriptive system-level contrast. Nevertheless, the completed configurations illustrate that greater policy control over action support need not coincide with lower recurrence: strategy support, temporal organization, and wording must be considered jointly.

This finding has practical implications for how designers configure interactive agents. The trade-off between alignment and repetition—between faithfully reproducing human action distributions and avoiding conversational ruts—cannot be resolved by simply granting the policy more or less control. Rather, it requires careful consideration of where in the generation pipeline different constraints should operate. The explicit action boundary provides a concrete substrate for such structural traceability. It enables system designers to compare the strategy selected for a state with the utterance ultimately produced, inspect recurring control patterns, and localize whether a divergence originates in policy choice or realization. We establish this as a method-level traceability capability. Nevertheless, for HCI researchers and practitioners building interactive systems, this form of inspectability is a prerequisite for responsible iteration and improvement.

5.4 Implications for Evaluating Interactive Dialogue Agents

Three implications follow from the present testbed for evaluating interactive agents that aim to engage in richer, more human-like dialogue. First, action-pattern metrics should be paired with controls whose inductive biases match those metrics; otherwise, local statistical reproduction may be mistaken for context-sensitive competence. An agent that reproduces the local distribution of human interaction moves may still fail to respond meaningfully to what has actually been said, a failure that becomes visible only when one intervenes on the dialogue content itself. Second, evaluation should treat strategy choice and language realization as separate but coupled objects: an appropriate action label is of little use when the utterance fails to express it. Whether the agent intends to guide, challenge, or acknowledge and whether it actually does so through language is an empirical question, not a given. Third, long-horizon repetition must

be reported alongside diversity and alignment, because a policy can cover human-supported patterns while repeatedly cycling among a small subset. For interactive agents deployed in sustained use, this is precisely the kind of failure that erodes user trust and engagement even when individual responses appear appropriate. Recent analysis of authentic student-LLM use likewise shows why evaluation must reflect how people actually interact with systems rather than only how a system is intended to be used, and why turn-level and whole-dialogue views are complementary [19].

Taken together, these findings suggest an evaluation framework for strategy-rich interactive agents: expose a readable strategy layer, assess whether the policy uses interaction context, and evaluate human-pattern alignment, temporal organization, and language realization as complementary rather than interchangeable properties. This framework speaks directly to the HCI community’s interest in designing systems that are not merely capable but genuinely usable and engaging in real human interaction. The action-mediated approach provides the transparency necessary to shift evaluation from the question of whether an agent produces fluent responses toward whether it interacts effectively and whether the basis for that interaction is observable.

6 Conclusion

Interactive agents are increasingly embedded in daily life, and dialogue remains the primary channel through which they serve human needs. Yet the gap between answering questions well and engaging in genuinely collaborative, adaptive interaction remains wide. This paper introduced an action-mediated formulation for learning human-grounded interaction strategies and instantiated it in LA-IQL. From fixed human interaction trajectories, LA-IQL learns an explicit policy over readable interaction acts through rationale-conditioned action prediction and offline inverse Q-learning, while keeping strategy selection separate from language realization. This formulation makes interaction decisions, temporal organization, and surface realization independently inspectable. This is a necessary condition for building agents that move beyond mechanical responsiveness toward natural, context-sensitive engagement.

Across held-out long-horizon classroom rollouts, the evaluated TD-regularized checkpoint produced less recurrent behavior while preserving similar local human-pattern alignment. In the evaluated inference configurations, varying how control is allocated between the policy and the role model revealed a trade-off: direct selection better preserved held-out human patterns, whereas proposal-constrained re-ranking reduced recurrence. Together, these findings support the claim that action mediation can reveal consequential differences in interaction organization that fluent text or local similarity alone can obscure.

The broader significance of this work lies in its treatment of human interaction as a learnable and inspectable resource. By making strategy explicit, we enable designers and researchers to pose sharper questions about what interactive agents have actually learned from human demonstration, how they organize behavior over time, and where their limitations reside. The evidence supports an inspectable modeling and evaluation interface for strategy-rich interactive agents. It is bounded to the evaluated checkpoints, structural behavior, and short excerpt-level judgments. These boundaries define the trajectory of future work: extending action-mediated learning to diverse interactional domains, validating structural properties in sustained user experience, and ultimately building interactive agents that do not merely respond to people, but engage with them.

Limitations

LA-IQL relies on a fixed interaction-act taxonomy derived from one classroom annotation scheme. The resulting 31-action space may omit pedagogically meaningful behaviors and may not transfer unchanged across subjects, grade levels, languages, cultures, or classroom formats. Although the framework is designed around domain-specific state

and action spaces and is motivated by a broader interaction-learning problem, this paper evaluates only its classroom-dialogue instantiation and makes no empirical transfer claim to other interactive-agent domains, including conversational assistants, multi-agent systems, or embodied agents. Because the method learns from offline expert demonstrations, it can reproduce annotation errors and systematic biases in the source corpus and cannot identify effective strategies that are absent from that corpus. The finite-context policy observation may miss long-range instructional goals even though the role model receives a longer dialogue history.

The completed Direct and Candidate configurations differ in rationale sampling temperature in addition to their action-selection mechanism. Their comparison therefore identifies a reproducible system-level trade-off but does not isolate the causal effect of decoding mode; decoding-specific causal attribution is outside the claims of this paper.

Generated rationales should not be interpreted as faithful causal explanations of the model’s internal computation. They are supervised textual intermediates and can be plausible while remaining incomplete or inconsistent with the selected action. The explored proposal-constrained adjustment is limited by candidate coverage: a strong Q-function cannot select an action absent from the proposal set, and it does not distinguish utterances sharing the same action label. Direct decoding avoids this support restriction but delegates realization of its selected action to a separate role model, which may fail to express that action faithfully. Finally, automatic action-distribution metrics do not establish educational effectiveness, and this paper makes no claims about learning outcomes or teacher-training utility. The supplementary human evaluation uses a small convenience sample, six same-topic excerpt sets, and a shortened instrument whose A/B/C positions are not fully balanced. It supports perceived variety and discussion support for the two generated systems, but does not establish sustained coherence, engagement, trust, preference, educational usefulness, or learning outcomes. Authentic transcript ratings are not used for model-comparison inference because transcription loss and unmatched excerpt length and dialogue state make them a non-equivalent reference condition. The RQ4 knowledge-graph endpoints likewise measure structures expressed in generated text, not changes in student cognition, factual correctness, instructional quality, or learning outcomes. Machine-checkable schema and utterance references do not by themselves establish that extracted entities and relations are semantically correct, and exact edge identities may vary across repeated automatic extraction. Moreover, the current alignment endpoints emphasize local action n -grams and transition distributions. A policy such as the first-order Markov control can score highly by conservatively reusing training-supported transitions without responding to the semantic content of the current classroom state. Our 50-state intervention provides evidence that the Full policy distribution changes with utterance semantics, but its deterministic donor and small sample bound the conclusion to distributional sensitivity rather than contextual decision quality.

Ethical Considerations

Classroom dialogue can contain personal information and involve minors, making privacy, consent, and data governance central to this work. We use the classroom data in accordance with applicable local regulations. Real identifying information was removed from the transcripts before use, and we will not release the original audio, video, or transcript data. Adult questionnaire volunteers were informed of the study purpose and data use before participating voluntarily. Their responses are analyzed in de-identified form and reported only in aggregate. Generated dialogue must not be presented as a substitute for qualified teachers or used for high-stakes assessment without human oversight.

An interactive agent built with this framework may reproduce cultural, linguistic, or pedagogical biases present in the demonstrations and in the underlying language models. In particular, a policy trained on a narrow classroom population may incorrectly treat one discourse style as universally expert. Deployment should therefore include subgroup

and cross-context evaluation, educator review, clear disclosure that outputs are machine-generated, and mechanisms for users to contest or override generated recommendations. LLM-generated rationale annotations also require quality auditing because fluent explanations can conceal labeling errors or unsupported assumptions.

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Symbol	Operational definition	Teacher example	Student example
Ipk	Prior-known-knowledge question: solicits a standard-answer fact, concept, rule, or method accessible through recall.	"Which coordinate is zero for a point on the y-axis?"	"Why are some tree rings wide and others narrow?"
Pk	Prior-known-knowledge response: supplies a recalled fact, concept, rule, or basic method with little further inference.	"The water tap's motion is rotational."	"Corresponding angles of similar triangles are equal."
Ipi	Personal-information question: solicits a subjective view, feeling, or interpretation grounded in personal experience.	"How else could you understand it?"	"What did that make you think of?"
Pi	Personal-information response: expresses a subjective view, feeling, or interpretation without requiring objective evidence.	"I think he went from one mountain to another."	"I want to thank the people who protect animals."
Ia	Analysis question: requests explanation, analysis, evaluation, or evidence beyond simple recall.	"Which words in the passage support that reading?"	"How can you be sure it has this shape?"
An	Analysis response: provides explanation, analysis, evaluation, or elaborated evidence.	"The particles are constantly moving."	"Because the tree can withstand life's trials."
Ic	Coordination question: asks a participant to summarize, compare, connect information, or identify a general pattern.	"Can you summarize it in one sentence?"	"So the zero mark must touch the measured object?"
Co	Coordination response: summarizes, compares, connects information, or states a general pattern.	"The text describes its appearance, activities, and habitat."	"Thus, the sounds of all four seasons are beautiful."
Is	Speculation question: asks a participant to infer, transfer, predict, or explore an unknown case from known information.	"Would the same pattern hold for a cart on a slope?"	"What if the triangle were changed to an equilateral one?"
Sp	Speculation response: offers an inference, transfer, prediction, or exploration of an unknown case from known information.	"Do not light fires in a forest; they can cause a wildfire."	"I also saw a small boat moving quickly."
Iu	Uptake question: invites further discussion, extension, deeper analysis, or an alternative view of a preceding contribution.	"Does anyone in your group want to add something?"	"What else could be added?"
Up	Uptake response: extends, analyzes, or offers an alternative to a preceding contribution.	"That is because it is a distinctive feature of Venice."	"The cart moves faster and faster down the slope."
Ag	Agreement: explicitly accepts, supports, or positively acknowledges a contribution.	"Good. Are there any other observations?"	"Yes—his high status made that possible."
Qu	Querying: disagrees with, doubts, corrects, or asks for reconsideration of a contribution.	"That is not precise enough; please try again."	"No angle is defined as right, so the conclusion does not follow."
Gu	Guide: gives an instruction, direction, prompt, or requested next step.	"Please verify whether it is this angle."	"Please practise this on your own."

Table 9. Operational definitions and role-specific examples for the 15 cognitive-act symbols. Crossing these symbols with teacher and student roles, and adding END, yields the 31-action policy space. Examples are translated and lightly abridged from authentic source transcripts and are illustrative rather than normative.

A Additional Experimental Results

A.1 Operational Action Definitions and Examples

Table 9 expands the compact vocabulary in Table 1. The examples are translated and lightly abridged from de-identified utterances in the source corpus; they illustrate how the same symbol can be performed by either role. Because annotation is assigned at the utterance level, an example may also contain conversational material outside the focal act.

A.2 Additional Quantitative Results

Training and inference configurations. LA-IQL directly selects the highest-valued action and then obtains one role-model realization. Candidate re-ranking instead generates six action-utterance proposals and constrains the policy to actions represented in that set. This changes the allocation of control between the policy and role model. It does not retrain the policy. The completed configurations also use different rationale temperatures (0.0 vs. 0.6), so the comparison below is a descriptive system-level contrast rather than an isolated decoding ablation. The w/o-TD Direct condition instead removes only the temporal-difference term from training and retains the main Direct generation configuration, including rationale temperature 0.0. Five of its 30 trajectories terminate after the minimum allowed horizon, compared with none for main Direct.

Split	Lessons	Turns	Topics	Transitions
Train	35	6,352	33	6,317
Validation	5	1,039	5	1,034
Test	10	2,028	10	2,018
Total	50	9,419	47	9,369

Table 10. Corpus statistics for the frozen lesson-level split. Topic counts within rows are distinct counts and need not sum to the corpus total because one topic name occurs in more than one partition.

Variant	D2 ↑	Cycle-2 ↓	Motif ↓	JSD ↓
LA-IQL w/o TD (Direct)	0.4118	0.2607	0.5379	0.0835
LA-IQL Direct	0.4370	0.1708	0.3957	0.0827
Candidate re-ranking	0.4478	0.0488	0.1413	0.1059

Table 11. Checkpoint and inference-time system comparisons on the same held-out lessons. The w/o-TD row is a fixed-checkpoint training-objective contrast; Candidate re-ranking reuses the Full checkpoint and is a descriptive system configuration, not a separately trained baseline. JSD is the paired lesson-level bigram divergence.

Method	State F1	Masked R-only F1	Combined F1	Δ_{sim}	Accuracy
LA-IQL w/o TD (Direct)	0.0257	0.3682	0.3629	0.3372	0.6867
LA-IQL Direct	0.0159	0.3986	0.4044	0.3885	0.7097
Candidate re-ranking	0.0123	0.1013	0.0986	0.0863	0.2393

Table 12. RQ3 rationale–action simulatability. Values are means of the three generation-seed-level scores. The target is each system’s selected action; the result is an association diagnostic, not a causal-faithfulness measure.

Figures 5 and 6 provide complementary visual summaries of these comparisons. The first emphasizes paired uncertainty, whereas the second preserves the reported metric scales and generation-seed-level variation.

RQ3 rationale–action simulatability. The independent predictor masks action symbols, canonical names, abbreviations, and deterministic label strings. State and rationale use separate character 2–5-gram TF–IDF representations and validation-selected linear classifiers. All three systems use the same frozen evaluator protocol.

Knowledge-graph evaluation coverage. The formal RQ4 extraction contains 250 valid graphs across nine systems, and all seven prespecified Direct–comparison families have matched lesson-cluster statistics. The reported evidence is restricted to the three graph endpoints in Table 8; no composite classroom-quality score is constructed.

Complete diversity and recurrence diagnostics. Table 13 reports the full diversity and recurrence suite for every completed system. Cov.@20 is the fraction of the 20 expert bigram or trigram patterns with greatest lesson support (with occurrence count as a tie-breaker) that appears in a generated run. These diagnostics must be interpreted jointly: high diversity or low recurrence alone does not imply expert-pattern alignment.

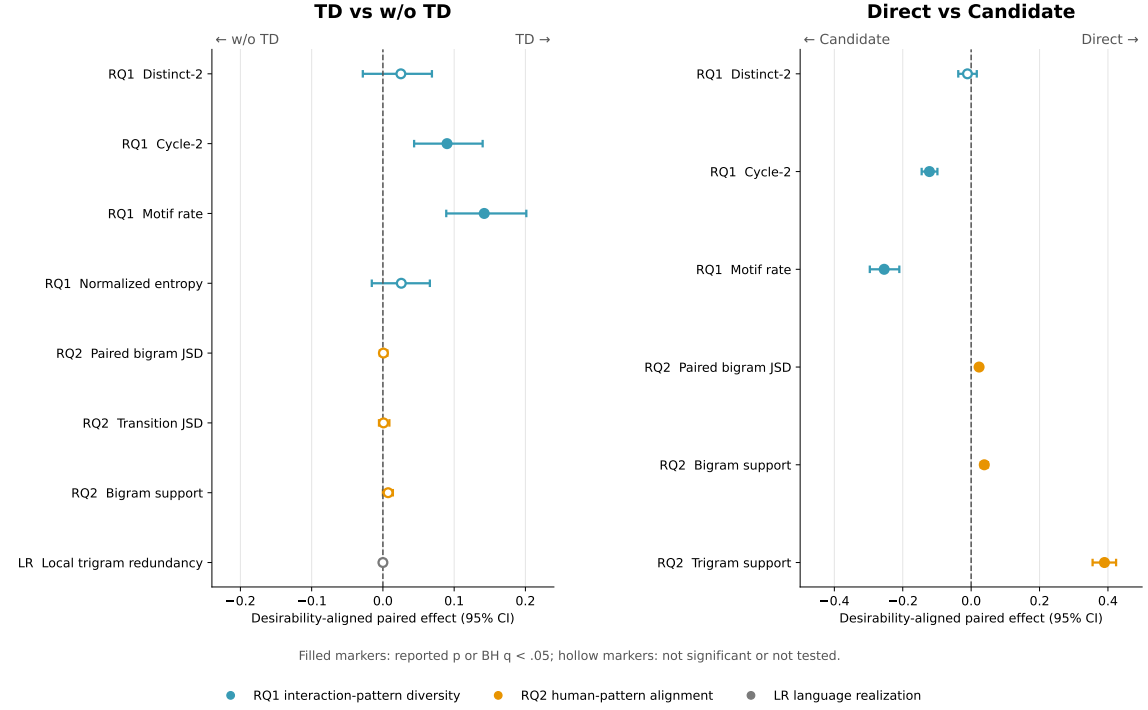


Fig. 5. Desirability-aligned paired effects for the fixed-checkpoint TD contrast and Direct-Candidate system contrast. For metrics marked ↓ in the result tables, the reported difference and interval are multiplied by -1 so that positive values consistently favor the first named system (TD or Direct). Horizontal bars are percentile 95% intervals from 10,000 paired bootstrap resamples clustered by the 10 held-out lessons. Filled markers denote a reported $p < .05$ or BH-adjusted $q < .05$; hollow markers denote a non-significant or untested comparison. The Direct-Candidate panel is a descriptive system-level comparison because rationale temperature also differs between configurations. LR marks the auxiliary language-realization diagnostic rather than an evaluation goal.

System or control	Runs	Dist.-1	Dist.-2	Dist.-3	Motif ↓	Cycle-2 ↓	Cov.@20	H_{norm}
<i>Generation baselines</i>								
Direct Role LLM	10×3	0.2692 ± 0.0050	0.4364 ± 0.0348	0.5913 ± 0.0427	0.4486 ± 0.0287	0.0286 ± 0.0088	0.5500 ± 0.0408	0.6722 ± 0.0102
Constrained Prompt	10×3	0.2866 ± 0.0029	0.5279 ± 0.0117	0.7264 ± 0.0144	0.3403 ± 0.0204	0.1520 ± 0.0147	0.5333 ± 0.0236	0.6818 ± 0.0015
<i>Statistical diagnostic controls</i>								
Uniform Random	10×3	0.4149 ± 0.0198	0.9593 ± 0.0048	0.9986 ± 0.0013	0.0048 ± 0.0012	0.0000 ± 0.0000	0.3500 ± 0.0000	0.9287 ± 0.0066
Expert Frequency	10×3	0.2154 ± 0.0145	0.6951 ± 0.0277	0.9449 ± 0.0069	0.0795 ± 0.0098	0.0067 ± 0.0033	0.9500 ± 0.0000	0.7144 ± 0.0111
First-order Markov	10×3	0.2004 ± 0.0047	0.5144 ± 0.0072	0.7815 ± 0.0026	0.2136 ± 0.0417	0.0819 ± 0.0140	1.0000 ± 0.0000	0.7072 ± 0.0012
<i>Learned-policy comparisons</i>								
Frozen BC	10×3	0.0303 ± 0.0009	0.0694 ± 0.0062	0.1156 ± 0.0132	0.9690 ± 0.0245	0.0072 ± 0.0069	0.3167 ± 0.1434	0.2379 ± 0.0134
LoRA BC	10×3	0.0250 ± 0.0085	0.0539 ± 0.0267	0.0895 ± 0.0496	0.9697 ± 0.0173	0.0017 ± 0.0013	0.3333 ± 0.2055	0.1267 ± 0.0940
<i>LA-IQL checkpoint and inference configurations</i>								
LA-IQL w/o TD (Direct)	10×3	0.1606 ± 0.0088	0.4118 ± 0.0236	0.6352 ± 0.0219	0.5379 ± 0.0361	0.2607 ± 0.0193	1.0000 ± 0.0000	0.6412 ± 0.0120
LA-IQL	10×3	0.1663 ± 0.0034	0.4370 ± 0.0101	0.6983 ± 0.0050	0.3957 ± 0.0240	0.1708 ± 0.0126	1.0000 ± 0.0000	0.6670 ± 0.0084
LA-IQL + candidate re-ranking	10×3	0.1493 ± 0.0070	0.4478 ± 0.0205	0.7221 ± 0.0238	0.1413 ± 0.0017	0.0488 ± 0.0013	0.7000 ± 0.0408	0.6664 ± 0.0131

Table 13. Complete diversity and recurrence diagnostics for all completed systems. Values are mean ± population standard deviation across seeds 42, 43, and 44; each seed contains 10 held-out lessons. BC seeds index independent training runs and their matching generation runs, whereas LA-IQL configurations reuse one checkpoint per condition across generation seeds. Diagnostic controls are included to show that high diversity or low recurrence alone does not establish expert-pattern alignment; the row groups are not a single fairness-matched ranking.

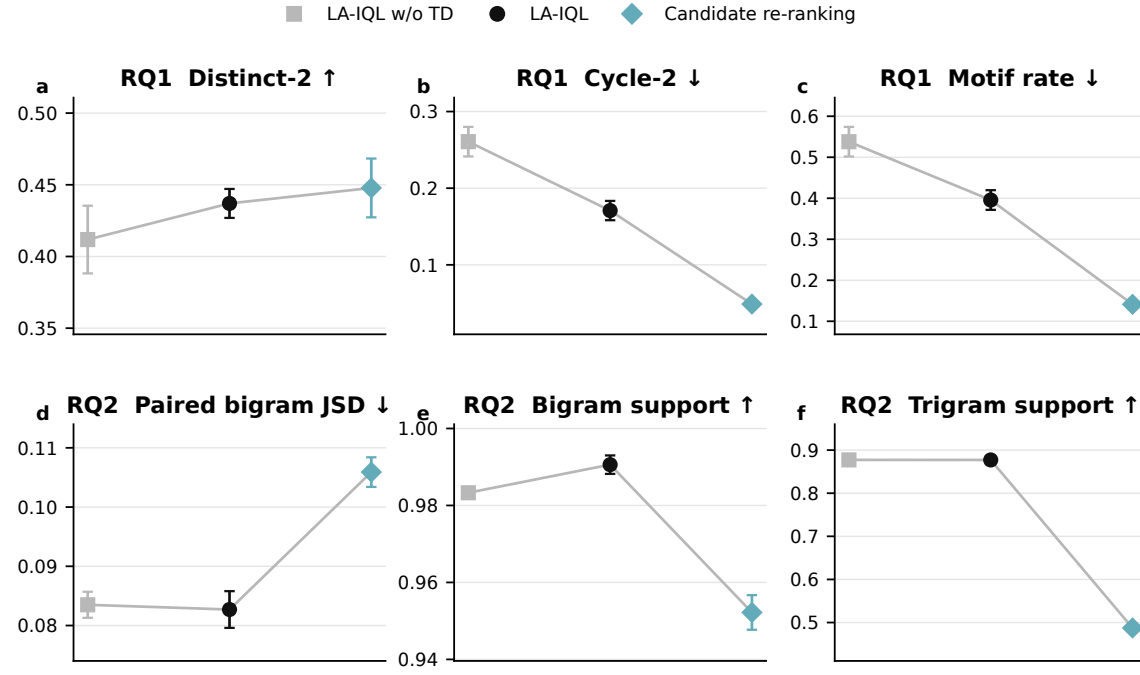


Fig. 6. Diversity and alignment results for the three LA-IQL configurations (RQ2). Points show means and error bars show population standard deviations across generation seeds 42–44 from each fixed checkpoint, with 10 held-out lessons per seed. Connecting lines are visual guides rather than temporal trajectories; arrows indicate the desirable metric direction.

B Implementation and Reproducibility

Archived training configuration and provenance boundary. We transcribed inference settings from the terminal config object in the formal trajectory JSON files, data provenance from the frozen split manifest, and model identity from checkpoint metadata. These artifacts are authoritative for the reported runs. The recovered run configuration confirms three two-epoch LoRA stages, the learning rates and loss weights below, physical batch size 4, one gradient-accumulation step, maximum length 512, gradient clipping at 1.0, and training seed 42. Full and no-TD share this configuration except that the Stage 3 TD-loss weight is 1 for Full and 0 for no-TD. The optimizer details, learning-rate scheduler and warm-up, LoRA rank/alpha/dropout and target modules, numerical precision, exact sample and optimizer-step counts, model/tokenizer revision, checkpoint-selection rule, and historical code revision were not retained in the archived artifacts and are therefore unavailable for this report. We do not substitute current defaults for those fields or describe the result as replicated across training seeds.

Across Stages 1–3, the recovery-loss weights are 0.5, 0.5, and 0.75; the action-token weights are 8, 5, and 5; the structure-token weights are 10, 8, and 8; the rationale-end-token weights are 50, 30, and 30; and the action-boundary-token weights are 30, 20, and 20. The TD discount is 0.99. The rationale generator uses maximum 256 new tokens and temperature 0 during training. Configuration comments that describe micro-batch 1 with accumulation 4 are stale; the actual run parameters are physical batch 4 and accumulation 1.

Completed BC training metadata. The frozen-backbone BC checkpoints use Qwen/Qwen3.5-0.8B, 6,181 training samples, next-action cross-entropy, three epochs, batch size 4, learning rate 2×10^{-5} , and maximum length 512. The LoRA BC checkpoints use the same backbone, objective, and training set plus 999 validation samples; they use three epochs, batch size 32, learning rate 10^{-4} , maximum length 256, rank 8, bfloat16, and one gradient-accumulation step. Each family retains independent training seeds 42–44 and per-epoch metadata.

Inference configuration and quality gates. All reported Full, Candidate, and no-TD formal trajectories use a two-turn, two-action policy window; a 30-turn, five-action role window; policy maximum length 512; rationale budget 256 new tokens; horizon 100; role temperature 0.8; three candidates per role where candidate re-ranking applies; and END suppression through step 49 with logit margin 3.0. The Full and no-TD generation gates each contain 50 samples and report 100% structural validity, closed rationale/action tags, no degeneration, and no length-limit hits. These checks establish schema validity, not policy or rationale quality. Frozen BC and LoRA BC use the same test lessons, horizon, END rule, role-model temperature, and role-model context; each selected BC action is realized by one role-model utterance. Their three seed-level runs pair independently trained checkpoints with matching generation seeds.

Software environment. The locked environment requires Python 3.10 or newer and resolves PyTorch 2.11.0+cu128, Transformers 5.14.1, PEFT 0.20.0, scikit-learn 1.7.2, SciPy 1.15.3, OpenAI 2.52.0, and Google GenAI 2.16.0. The dependency lockfile and CUDA 12.8 wheel index are part of the code artifact.

C Context-Sensitivity Intervention

The frozen context-sensitivity-v1 protocol selects five pre-existing, evenly spaced held-out states per lesson from the frozen RQ4 candidate pools. For each state, it constructs a within-lesson cyclic semantic donor while preserving the number and sequence of speakers and the complete action history. Rationales are greedily decoded with temperature 0 and a 160-token budget; END suppression is not applied because this is a state-level test. Intervals use 10,000 percentile bootstrap resamples clustered by lesson with seed 20260822.

Stage	Initialization	Epochs	LR	Policy	Cons.	LM-act	TD Full/no-TD	Teacher/self
1	Base model	2	10^{-4}	0.25	0	1.0	0/0	1.0/0.0
2	Stage 1 final	2	10^{-4}	1.0	0.1	0.5	0/0	0.8/0.2
3	Stage 2 final	2	5×10^{-5}	1.0	0.1	0.5	1/0	0.6/0.4

Table 14. Recovered three-stage configuration for the evaluated Full and no-TD checkpoints. “Cons.” is the cross-head consistency-loss weight and “LM-act” is the LM action-class loss weight. The two configurations differ only in the Stage 3 TD-loss weight among the recovered fields.

System	Checkpoint / objective	TD	Selection	Rat. temp.	Role proposals	Randomness
Full Direct	sft_stage3/final; four-term objective	on	full-space argmax	0.0	one realization	gen. 42–44
Full Candidate	same Full checkpoint	on	six-proposal re-rank	0.6	3/role	gen. 42–44
No-TD Direct	sft_stage3_woTD/final; TD removed	off	full-space argmax	0.0	one realization	gen. 42–44
Frozen BC	sft_bc_seed*; next-action CE	off	full-space argmax	n/a	one realization	train/gen. 42–44
LoRA BC	lora_policy_seed*; next-action CE	off	full-space argmax	n/a	one realization	train/gen. 42–44
LM likelihood rerank	Qwen/Qwen3.5-0.8B; no policy training	off	six-proposal LM rerank	n/a	3/role	gen. 42–44

Table 15. Learned-policy and reranking-system provenance matrix. “gen.” denotes generation seeds applied to a fixed checkpoint or inference procedure; “train/gen.” denotes independently trained BC checkpoints paired with the matching generation seed. Both BC systems are included in the formal RQ1/RQ2 action-trajectory comparisons and in the RQ4 knowledge-graph evaluation. The likelihood reranker is an inference-time baseline and has no trained policy checkpoint.

Endpoint	Original	Counterfactual	Difference / shift
Accuracy	0.4000	0.3000	−0.1000
Macro-F1 (31)	0.1486	0.1285	−0.0201
Expert-action probability	0.2732	0.1965	−0.0767
Prediction flip rate	—	—	0.6000
Policy JSD	—	—	0.2957

Table 16. Exploratory context intervention for the main Full checkpoint ($N = 50$ states, 10 lessons). Lesson-cluster intervals appear in the main text. The intervention tests distributional sensitivity, not contextual correctness.

D Evaluation Details

Additional diversity and recurrence metrics include Distinct-1/3, repeated local motifs, and cross-trajectory variance. For RQ3, masking removes action symbols, canonical names, abbreviations, and deterministic label strings before prediction. We report leakage rates, per-class scores, confusion matrices, and majority and shuffled-rationale controls.

The RQ3 evaluation provides association evidence through independent prediction and within-lesson/role shuffle controls. Its evidential scope is simulatability; it does not include causal interventions such as removal, paraphrase, or role-compatible counterfactuals [9], nor human judgments of action support or classroom appropriateness. Accordingly, this automatic RQ3 diagnostic does not establish explanation faithfulness, educational effectiveness, or experienced naturalness. The supplementary human evaluation separately reports excerpt-level perceptions.